RM Kummer FFT rounding guard correction

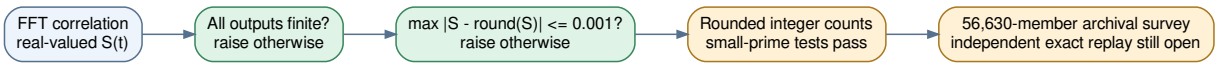
Implementation validation note

25 October 2026

Scope. Source snapshot `af671797`. No archived survey record or performance value is changed.

Observed discrepancy. The frozen survey README says the FFT point counter raises when its output is more than 10^{-3} from an integer. The previous code required that absolute condition *and* an additional relative condition. The synthetic value $S = 6000.003$ has residual 0.003 but passes the old guard because $0.003 < 10^{-6}|S|$. NaN and infinity also bypassed those comparisons. These are guard counterexamples, not observations from the archived survey.

Repair. Reject nonfinite output, then raise whenever $\max |S - \mathrm{round}(S)| > 10^{-3}$. The targeted regression injects all three counterexamples; the existing small-prime FFT versus brute-force order test remains a separate control. Targeted result: four passed, eleven deselected. The full RM/Q-curve test module passed fifteen of fifteen.



Boundary. This check does not certify the true integer character sum for each archived member. The 56,630-member RM survey, 5,628 split cases and 859 recorded hits have a separate summary-arithmetic check and 12 reported PARI spot checks, but no independent exact member-by-member replay or original loaded-source custody. No canonical measured graph changes.

An open problem record tracks the exact-census and source-custody obligations. Future generated survey prose now labels the FFT counts as rounded results and states the validation limit.

Reproduction. See `REPORT.md`, `guard.dot`, the changed `rmkummer.py`, and `tests/test_endosweep_rm_qcurves.py` in this branch. The diagram shows the corrected guard and the remaining archival-validation boundary.